

Player Coach Database with Chatbot

AN EXPERIENTIAL LEARNING PROJECT REPORT SUBMITTED TO
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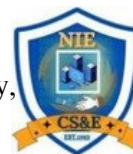
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CERTIFICATE

This is to certify that the Experiential Learning project work entitled "**Player-Coach database with Chatbot**" is a Bonafede work carried out by Sathwik N H(4N122CI051), Shama G Hegde (4N122CI052), Shaunak Banerjee(4N122CI053), Sinchan M S (4N122CI054), Sourav S Avadhani(4N122CI055) in partial fulfillment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering(AIML)**, of Visvesvaraya Technological University, Belagavi, during the year 2024-2025. It is certified that all suggestions indicated during internal assessment have been incorporated and the corrected copy has been deposited in the department library. This project report has been approved in partial fulfillment for the award of the said degree as per academic regulations of The National Institute of Engineering.

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Chapter 1

INTRODUCTION

AI-driven assistant systems have become a significant part of modern digital platforms, offering users intuitive and intelligent ways to interact with information. In the context of sports and team management, these systems hold the potential to transform how coaches plan, manage, and make decisions. By combining artificial intelligence and natural language processing, a coach assistant platform can provide real-time insights, strategic suggestions, and efficient team management—all through a simple, conversational interface.

Background: In the evolving landscape of sports technology, traditional coaching methods often rely on manual data tracking and intuition-based decisions. This approach can be time-consuming and lacks the dynamic, data-driven capabilities that modern tools can offer. With the rise of AI-powered systems, coaches can now leverage technology to make smarter, faster decisions. Inspired by this potential, the Coach Assistance Platform integrates an intelligent chatbot system capable of analyzing player information, suggesting training plans, diet strategies, and even game tactics based on team dynamics

By utilizing advanced language models such as Gemma 2-2B, the platform acts as a smart assistant for coaches—streamlining daily activities and enhancing overall team performance. This project represents a practical application of AI in sports management, bridging the gap between data analysis and real-time strategic support.

The purpose of this project is to develop an AI-powered Coach Assistance Platform that supports coaches in managing their teams more effectively. The platform is designed to display key player details—including position, height, weight, and skillsets—while providing coaches with strategic suggestions through a built-in chatbot. Coaches can interact with the assistant to receive training recommendations, diet plans, and match strategies tailored to their team composition.

This project demonstrates how AI and natural language processing can be harnessed to improve decision-making, enhance productivity, and simplify daily coaching tasks. It aims to create a more accessible and interactive experience for coaches, highlighting the real-world potential of integrating intelligent systems into sports and team management.

The Coach Assistance Platform aims to simplify the daily operations of a coach by providing instant access to player details, including key attributes like position, height, weight, and skills. The integrated AI chatbot offers personalized recommendations based on real-time data, such as training plans, diet strategies, and match tactics. The assistant can also help coaches with strategic decisions, offering suggestions on optimal formations, substitutions, and tactics for upcoming games, based on the analysis of player performance and team dynamics.

By automating routine tasks and offering AI-driven insights, the platform significantly reduces the time and effort coaches need to spend on manual data analysis. With instant feedback on player performance and real-time recommendations, coaches can make faster, more informed decisions that are tailored to their team's specific needs. This not only enhances team performance but also supports player development by creating personalized training and fitness plans. The platform provides coaches with a data-driven approach to decision-making, helping them stay ahead in competitive sports environments.

Looking ahead, the Coach Assistance Platform has the potential for further development to include more advanced features, such as predictive analytics, injury management, and deeper machine learning models for team strategy optimization. With future enhancements, the platform could provide deeper insights into player injuries, fatigue, and recovery, enabling coaches to manage their teams more effectively and safely. As AI technology continues to evolve, so too will the capabilities of this platform, providing coaches with even more powerful tools to enhance team management and performance.

Chapter 2

ABOUT THE PROJECT

The Coaching Assistant Platform is an innovative AI-driven application designed to streamline team management and enhance coaching strategies. Traditionally, coaches rely on manual data analysis, intuition, and spreadsheets to manage their teams and make decisions regarding training, match strategies, and player development. This approach, while effective, can be time-consuming and often lacks the dynamic, data-driven insights that modern tools can provide. With advancements in artificial intelligence, the Coaching Assistant Platform automates many of these tasks, empowering coaches with real-time, data-backed insights and recommendations.

Leveraging powerful AI models like Gemma 2-2B, this platform acts as a personal coaching assistant, capable of analyzing player performance, offering training suggestions, and optimizing match strategies—all through a conversational interface. Coaches can interact with the assistant to get detailed player information, suggest customized training and diet plans, and receive tactical advice for upcoming matches, all tailored to the specific needs of the team. The system is designed to help coaches focus on strategy and player development by automating time-consuming tasks like data collection and analysis.

The platform is built with user-friendliness in mind, offering a seamless experience for coaches to manage their teams and interact with the AI assistant. The backend is powered by robust technologies like Node.js, Express.js, and MongoDB, while the frontend leverages React for an intuitive, responsive interface. The chatbot interacts with the AI model in real-time to deliver actionable insights based on current team data, ensuring coaches can make faster and more informed decisions.

The Coaching Assistant Platform aims to revolutionize coaching by combining artificial intelligence with sports management, offering a practical and efficient solution for team optimization. It serves as a foundation for future advancements in sports AI, providing coaches with powerful tools to improve team performance, make data-driven decisions, and enhance the overall coaching experience.

The Coaching Assistant Platform has great potential for future growth, including features like predictive analytics, injury monitoring, and real-time game support. These upgrades will help coaches make faster, data-driven decisions, optimize training, and improve team performance, making the platform an essential tool for modern coaching.

Chapter 3

SYSTEM DESIGN & TECH STACK

The Coaching Assistant Platform is a web-based application that integrates AI-powered insights with a user-friendly interface for team management and coaching support. Designed to be efficient, scalable, and intuitive, the platform enables coaches to make data-driven decisions, optimize player performance, and improve overall team management. This section provides an in-depth breakdown of the system architecture, components, and technologies used in the development of the platform.

System Design:

The Coaching Assistant Platform follows a client-server architecture, with a React-based frontend that communicates with a Node.js and Express.js backend via RESTful APIs. The backend handles user authentication, data processing, and AI model interaction, while the frontend offers a responsive and interactive interface for coaches to manage their teams and access AI-generated recommendations.

- Built using React.js to provide a dynamic and responsive interface.
- Coaches can log in and view a dashboard displaying player details such as name, position, height, weight, and skills.
- A chatbot interface is integrated in the bottom corner, allowing coaches to interact with the AI assistant.
- Designed using CSS and basic layout frameworks for improved user experience and visual clarity.

2. Backend (API Server)

- Developed using Node.js with Express.js to handle server-side operations.
- Manages user authentication, session handling, and access control for coaches.
- Exposes RESTful API endpoints for login, retrieving player data, and sending user queries to the AI model.
- Handles communication between the frontend and the AI model.

3. AI Model Integration

- Integrates with Gemma 2-2B via a separate AI interface or external service.
- Backend receives queries from the chatbot interface and forwards them to the AI model.
- The AI model generates context-aware responses (e.g., strategy suggestions, training advice) and returns them to the coach through the frontend.

4. API Communication Layer

- The React frontend makes HTTP requests to the Express backend via REST APIs.
- The backend fetches data from the MongoDB database or sends queries to the AI model as needed.
- Coaches receive AI responses and player information in near real-time through the inter

5. Database (MongoDB)

- MongoDB is used to store and retrieve structured data like user accounts, player profiles, and team data.
- Each coach's login allows access only to their associated player records.

Technology Stack:

The system is developed using a combination of modern tools and technologies to ensure efficiency, scalability, and ease of development.

1. Programming Languages

- JavaScript – Used for both frontend and backend development to ensure consistency across the application.
- HTML & CSS – Used to design and style the web interface, ensuring a clean and responsive user experience.

2. Web Framework

- Node.js and Express.js – Used for backend development to handle user authentication, API requests, and data processing efficiently.

3. AI Models & APIs

- Gemma 2-2B – A powerful language model used to provide coaching insights such as training suggestions, game strategies, and dietary recommendations.
- The backend processes queries from the frontend (e.g., via chatbot interface), sends them to the AI model, and returns contextually relevant responses .

4. API Communication

- The React frontend interacts with the backend via HTTP API requests for tasks such as:
- User login and authentication
- Fetching and displaying player details
- Submitting coaching queries to the AI model
- The backend handles all communication with the AI engine and delivers responses to the frontend seamlessly.

5. Frontend Technologies

- React.js – Offers a fast, interactive, and modular UI
- CSS – Ensures clean and responsive styling across devices.

6. Database (MongoDB)

- MongoDB – A flexible NoSQL database used to store user accounts, player profiles, and related team data.
- Coaches can access only their associated data, ensuring data privacy and personalization.

Chapter 4

Workflow & Data Flow

1. User Input Stage

- The coach logs into the platform and enters team-related information, such as player details (position, height, weight, skills) or requests AI-driven recommendations for training, match tactics, or diet plans.
- The frontend sends the input data as an API request to the Node.js Express backend.

2. Backend Processing Stage

- The backend receives and validates the input data, ensuring all necessary fields are present (e.g., player data or team preferences).
- Based on the coach's request (e.g., tactical advice, training plans), the backend constructs an appropriate prompt to send to the AI model.
- If the request involves personalized insights or recommendations, the backend prepares the data in the required format before calling the AI model.

3. AI Model Interaction

- The backend sends the constructed prompt to the Gemma 2-2B AI model (via API calls).
- The AI model processes the prompt, analyzes the input data (e.g., player stats, team performance), and generates insights, such as recommended strategies, training schedules, or diet plans.

4. Response Handling & Display

- Once the AI model processes the request, the backend formats the response into a structured, user-friendly format (e.g., JSON).
- The frontend receives the response and dynamically renders the content—showing the coach personalized insights, training recommendations, or match strategies on the user interface.

5. Logging & Error Handling

- The backend logs errors and exceptions for debugging and performance monitoring.
- If any AI model or API call fails (e.g., server unavailability), the system has a fallback mechanism to handle the failure and ensure continuous operation (e.g., default responses or error messages). If necessary, it could switch to a secondary model or notify the coach of the issue.

Chapter 5

Future Enhancements

1. Cloud Storage & Export Options

- Enable saving team/player data and AI-generated recommendations to the cloud for easy access and backup.
- Provide export options (CSV, PDF, or custom reports) for player data and coaching strategies.

2. Speech & Voice Generation

- Implement text-to-speech (TTS) models for AI-generated recommendations and strategies, allowing coaches to receive verbal feedback.

3. Advanced Player & Team Customization

- Allow coaches to specify detailed player attributes (e.g., fitness levels, preferred positions) and customize team strategies more precisely.

4. Real-Time Collaboration

- Develop a multi-user workspace for coaches to collaborate, share strategies, and provide feedback on team management and player development.

5. Predictive Analytics & AI-Driven Forecasting

- Integrate predictive analytics to forecast player performance, potential injuries, and match outcomes based on historical data.

6. Mobile Application

- Develop a mobile version of the platform for coaches to access team data, strategies, and AI recommendations on the go.

Chapter 6

IMPLEMENTATION

Pseudocode

```
iimport express from "express";
import mongoose from "mongoose";
import dotenv from "dotenv";
import cors from "cors";
import authRoutes from "./routes/authRoutes.js";
import playerRoutes from "./routes/playerRoutes.js";
import coachRoutes from "./routes/coachRoutes.js";
import chatbotRoutes from "./routes/chatbotRoutes.js"; // New Chatbot Route
import Coach from "./models/Coach.js";
import Player from "./models/Player.js";
import bcrypt from "bcryptjs";
import axios from "axios";

dotenv.config();
const app = express();
app.use(express.json());

// Configure CORS Properly
const corsOptions = {
  origin: "mport os import logging from typing import Optional from fastapi import FastAPI, Request, Form, HTTPException from fastapi.templating import Jinja2Templates from fastapi.staticfiles import StaticFiles from fastapi.responses import HTMLResponse import httpx from pydantic import BaseModel, Field from prompt_templates import SCENE CHARACTER PROMPT", // Allow frontend to access backend
  credentials: true, // Allow authentication headers & cookies
  methods: "GET,HEAD,PUT,PATCH,POST,DELETE",
};
app.use(cors(corsOptions));
```

```
// MongoDB Connection
const connectDB = async () => {
  try {
    await mongoose.connect(process.env.MONGO_URI, {
      useNewUrlParser: true,
      useUnifiedTopology: true,
    });
    console.log(" MongoDB Connected");
  } catch (error) {
    console.error(` MongoDB Connection Error: ${error.message} `);
    process.exit(1);
  }
};
connectDB();

// Insert Static Coaches & Players (Runs Once)
const insertStaticData = async () => {
  try {
    await Coach.deleteMany();
    await Player.deleteMany();

    const hashedPassword = await bcrypt.hash("password123", 10); // Securely hash passwords

    const coaches = [
      { name: "Coach A", email: "coachA@example.com", password: hashedPassword },
      { name: "Coach B", email: "coachB@example.com", password: hashedPassword },
      { name: "Coach C", email: "coachC@example.com", password: hashedPassword },
    ];
    const insertedCoaches = await Coach.insertMany(coaches);

    const players = [
      { name: "Player 1", position: "Forward", height: 180, weight: 75, coach: insertedCoaches[0]._id },
      { name: "Player 2", position: "Midfielder", height: 175, weight: 72, coach: insertedCoaches[1]._id },
    ],
    { name: "Player 3", position: "Defender", height: 185, weight: 80, coach: insertedCoaches[2]._id },
    ];
    await Player.insertMany(players);
    console.log(" Static Coaches and Players added successfully!");
  } catch (error) {
    console.error(" Error inserting static data:", error);
  }
}
```

```
insertStaticData();
```

```
// API Routes
```

```
app.use("/api/auth", authRoutes);
```

```
app.use("/api/players", playerRoutes);
```

```
app.use("/api/coaches", coachRoutes);
```

```
app.use("/api/chatbot", chatbotRoutes); // Chatbot endpoint (does not require playerId)
```

```
// Root Endpoint
```

```
app.get("/", (req, res) => res.send("API is running..."));
```

```
// API to Update Player Tactics
```

```
app.post("/api/update-tactics", async (req, res) => {
```

```
  try {
```

```
    const { playerId, newTactics } = req.body;
```

```
    if (!playerId || !newTactics) {
```

```
      return res.status(400).json({ error: "Player ID and new tactics are required" });
```

```
    }
```

```
    await Player.findByIdAndUpdate(playerId, { tactics: newTactics });
```

```
    res.json({ message: "Tactics updated successfully" });
```

```
  } catch (error) {
```

```
    console.error("Error updating tactics:", error);
```

```
    res.status(500).json({ error: "Failed to update tactics" });
```

```
  }
```

```
});
```

```
// Start Server
```

```
const PORT = process.env.PORT || 5000;
```

```
app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
```

Chapter 7

SCREENSHOTS

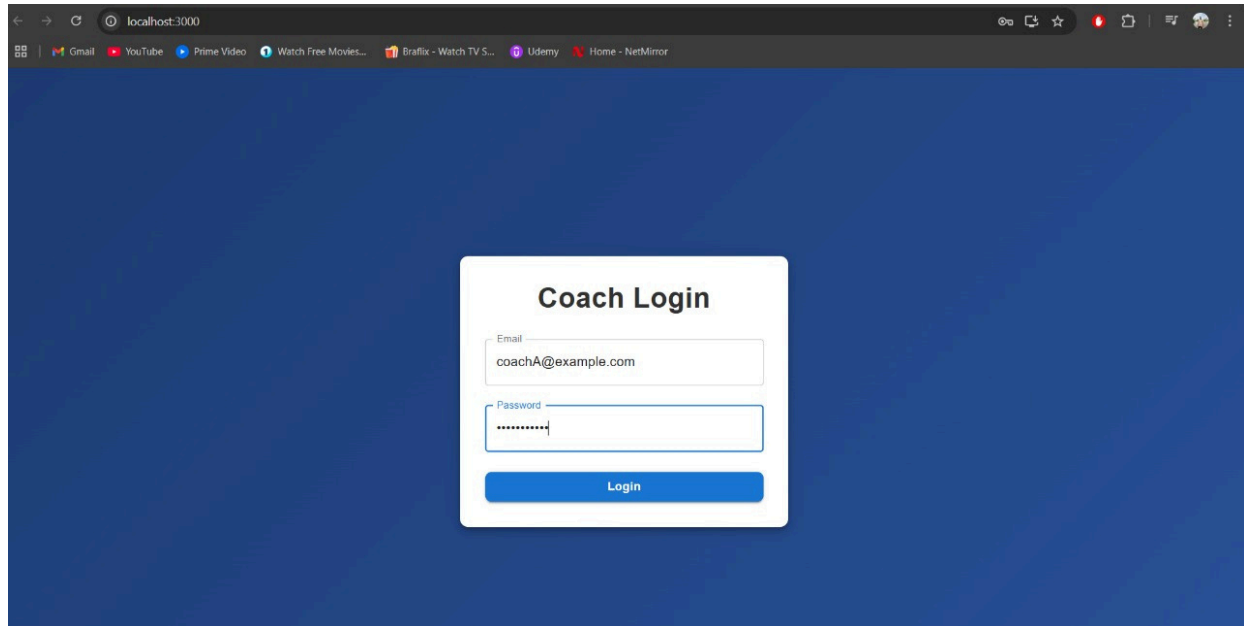


Fig 7.1

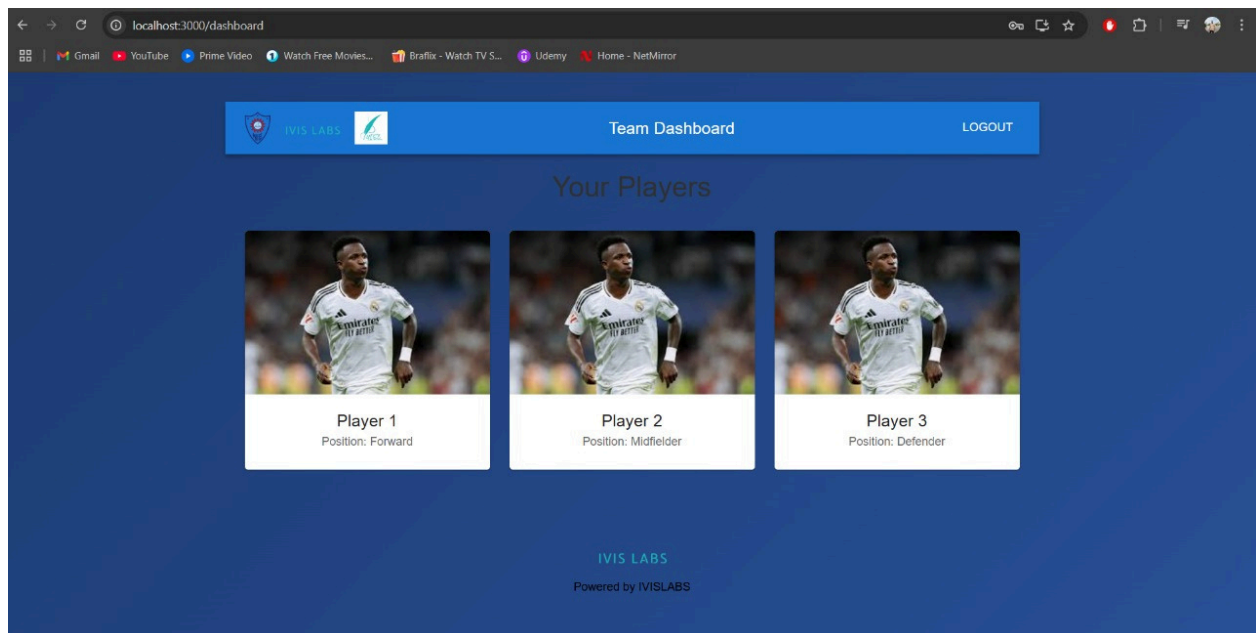


Fig 7.2

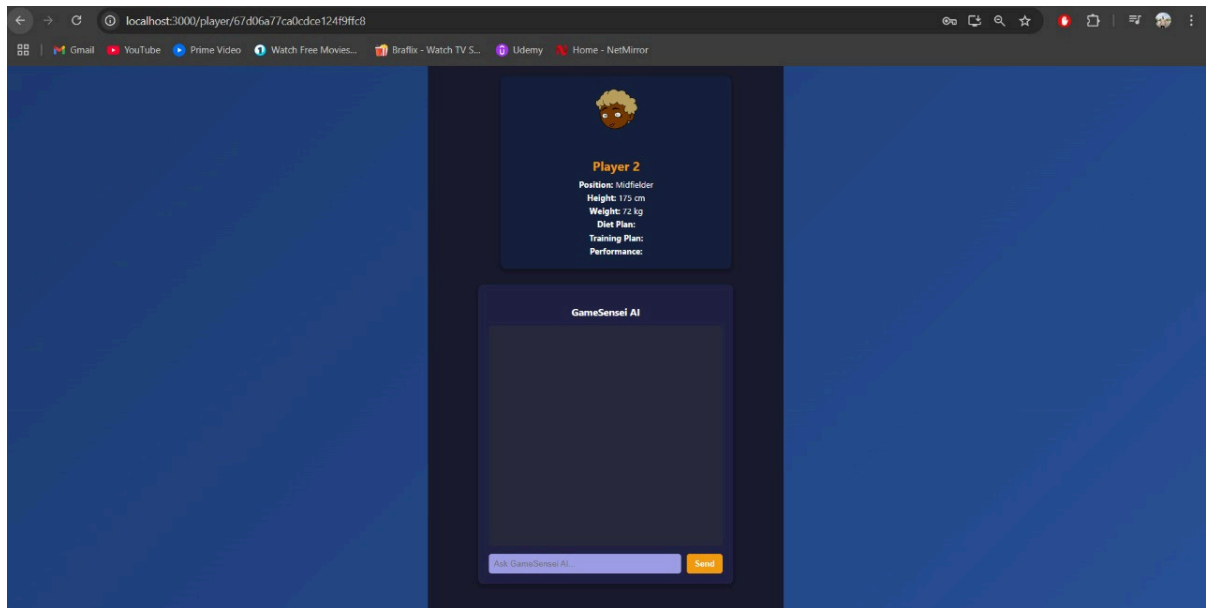


Fig 7.3

CONCLUSION

The Coaching Assistant Platform is a transformative AI-driven tool that enhances team management and coaching strategies by providing real-time insights and personalized recommendations. Traditional coaching methods often rely on manual tracking and intuition, which can be time-consuming and prone to errors. This project demonstrates how artificial intelligence can bridge this gap by offering data-driven suggestions for training, nutrition, and match tactics, tailored to the specific needs of a team and its players.

By integrating advanced AI models like Gemma 2-2B, the platform provides coaches with smart, actionable recommendations that enhance decision-making and optimize team performance. Whether it's developing personalized training plans, suggesting dietary strategies, or offering tactical advice, the system ensures that coaches have a comprehensive toolset to improve their teams' results. The use of a user-friendly web interface allows easy access to these insights, helping coaches at all levels to make informed choices

Additionally, the system's flexible architecture and error-handling mechanisms ensure continuous service, even if certain components or external APIs become temporarily unavailable. This is critical for coaches relying on the platform during crucial moments, such as game preparation or training sessions.

Beyond its technical accomplishments, the Coaching Assistant Platform emphasizes the growing role of artificial intelligence in sports and team management. Just as AI is revolutionizing industries like healthcare and finance, it is also transforming the way coaches manage their teams. This project serves as an example of how AI can empower coaches, making them more efficient and effective in guiding their teams to success, while also encouraging a data-driven approach to sports management.

BIBLIOGRAPHY

Books:

Graves, A. (2013). Speech Recognition and Artificial Intelligence. Springer.

This book explores speech recognition, a key component in chatbot interactions and provides insight into AI's application in real-time conversations.

Rossi, G., & Cohn, D. (2020). Data Science for Sports Analytics. Springer.

This text dives into the application of data science in sports, particularly in analyzing player performance, which could be a key feature in a Player Coach Database.

Articles and Papers:

Fink, J., & Bowers, A. (2019). "A Review of Chatbots in Sports and Their Applications." International Journal of Sports Science and Coaching, 14(4), 486-501.

This article reviews the use of chatbots in sports, detailing their applications in coaching, player performance tracking, and fan interaction.

Smith, M. & Brown, P. (2022). "AI in Sports: Improving Coaching through Technology." Journal of Artificial Intelligence in Sports, 3(2), 77-91.

This paper discusses how AI and machine learning models are being used to assist coaches with player performance data analysis and how chatbots can support coaching staff.

Johnson, L., & Thompson, E. (2020). "Building a Sports Player Database with Machine Learning." Sports Analytics Journal, 5(1), 34-56.

Focuses on creating databases to track player statistics and performance metrics, a foundational component of any Player Coach Database.

Online Sources:

OpenAI. (2025). Using GPT Models in Sports Analytics and Coaching. Retrieved from <https://openai.com>

OpenAI's documentation on using GPT-3 and GPT-4 models for creating advanced chatbot solutions in various industries, including sports.

Kaggle. (2023). Player Performance Data for Sports Analytics. Retrieved from <https://www.kaggle.com>

A dataset repository containing player statistics, used for building sports databases and analyzing player performance, key for a Player Coach Database project.